Project Design Phase
Solution Architecture

Date	04 April 2026
Team ID	SWTID-2026-2136
Project Name	Automating Employee Data
Maximum Marks	4 Marks

Solution Architecture

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram

Figure 1 below illustrates the end-to-end data flow and component relationships for the Automating Employee Data solution within ServiceNow.

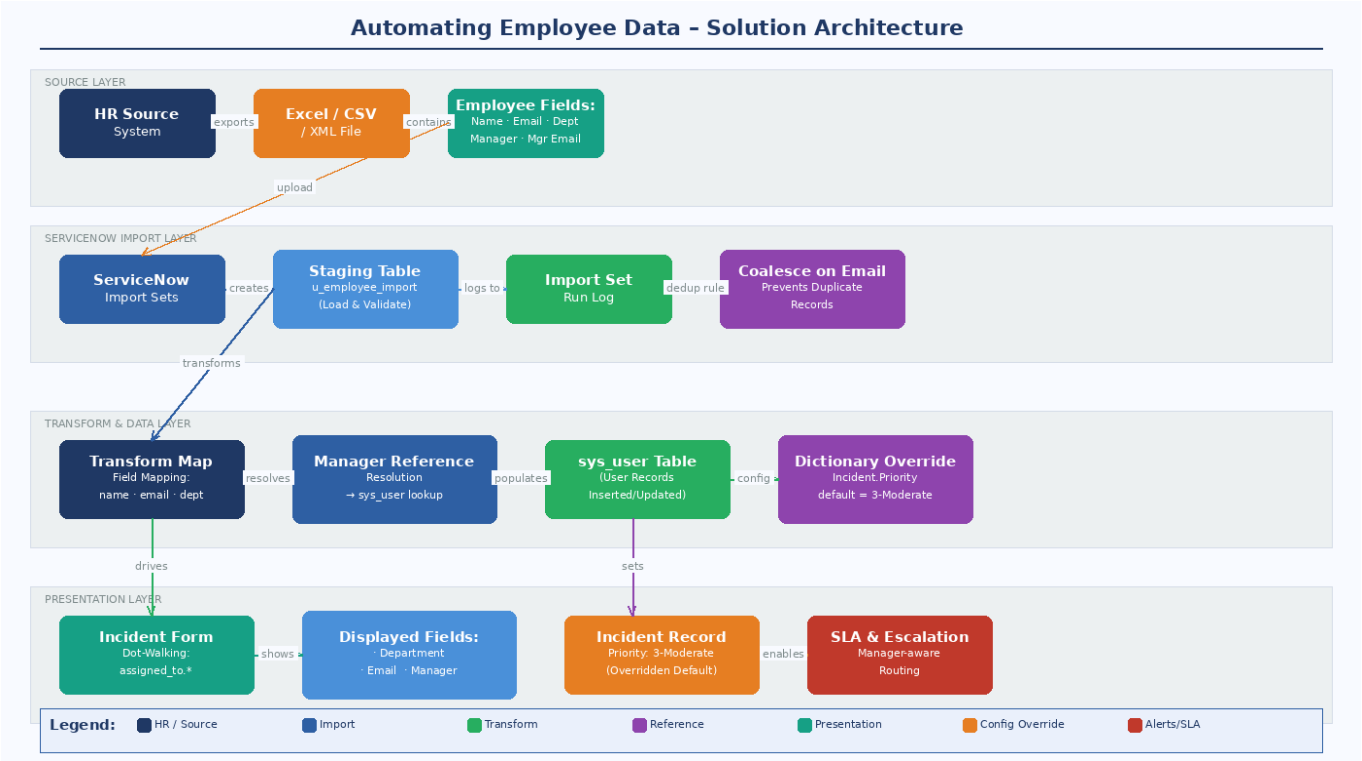

Figure 1: Solution Architecture and Data Flow – Automating Employee Data in ServiceNow

Architecture Component Description

Layer 1 – Source Layer

The source layer represents the origin of employee data before it enters the ServiceNow platform.

- HR Source System: The authoritative system of record (e.g. HRIS, Workday, BambooHR) from which employee data is exported periodically.
- External Data File: Employee records are exported in Excel, CSV, or XML format containing the five mandatory fields: Name, Email, Department, Manager, and Manager's Email.
- Data Preparation: HR administrators validate and clean the file, ensuring correct formatting, no missing required fields, and consistent naming conventions for manager lookup.

Layer 2 – ServiceNow Import Layer

This layer handles data ingestion into ServiceNow via the native Import Sets framework.

- ServiceNow Import Sets: The prepared file is uploaded through System Import Sets > Load Data, triggering the creation of an auto-generated staging table (e.g. u_employee_import).
- Staging Table: Acts as a temporary holding area where incoming data is stored and validated before transformation. Each column from the source file maps to a column in the staging table.
- Import Set Run Log: Records import metadata including row counts, errors, and warnings. Allows administrators to verify that all expected records were ingested before proceeding.
- Coalesce on Email: A coalesce rule on the email field is configured in the Transform Map to prevent duplicate sys_user records during repeated import runs.

Layer 3 – Transform & Data Layer

The transform layer converts staged data into validated ServiceNow user records, handling reference field resolution and table-specific configuration.

- Transform Map: Defines field-level mappings from the staging table to the sys_user table. Each source field is mapped to its target counterpart with appropriate data type handling.
- Manager Reference Resolution: The Manager field is treated as a reference lookup. The Transform Map resolves the manager name or email to an existing sys_user record, populating the manager reference field correctly.
- sys_user Table: The destination table for all transformed employee records. Upon successful transform execution, each employee row becomes a user record in ServiceNow with accurate department, email, and manager associations.
- Dictionary Override – Incident Priority: A Dictionary Override is applied to the Incident table's Priority field, setting a default value of 3-Moderate. This override is scoped exclusively to the Incident table and does not affect the parent Task table or sibling tables such as Change Requests or Problems.

Layer 4 – Presentation Layer

The presentation layer exposes transformed employee data within the ServiceNow Incident management interface.

- Incident Form – Dot Walking: The Incident form is configured with three additional fields that leverage dot walking through the assigned_to reference:
  - assigned_to.department – Displays the assignee's department directly on the form.
  - assigned_to.email – Displays the assignee's email without requiring a separate lookup.
  - assigned_to.manager – Displays the manager of the assigned user, enabling immediate escalation context.

- Incident Record with Overridden Priority: When a new Incident is created, the Priority field defaults to 3-Moderate as per the Dictionary Override, ensuring consistent prioritization across the organization.
- SLA & Escalation Routing: With manager information visible on the Incident form and accurate priority defaults, SLA tracking and manager-aware escalation routing are immediately effective upon go-live.

End-to-End Data Flow

Step	Component	Action / Output
1	**HR Source System**	Exports employee data as Excel/CSV/XML with Name, Email, Department, Manager, Manager Email.
2	**Import Sets – Load Data**	File uploaded to ServiceNow; staging table u_employee_import created; all rows loaded.
3	**Staging Table Validation**	Import Set Run Log reviewed; row count verified; data quality confirmed before transform.
4	**Transform Map Execution**	Field mappings applied; Manager resolved as sys_user reference; Coalesce prevents duplicates.
5	**sys_user Table Update**	Employee records inserted or updated in the User table with full field population.
6	**Incident Form Config**	Dot-walking fields added: assigned_to.department, assigned_to.email, assigned_to.manager.
7	**Dictionary Override**	Incident.Priority default set to 3-Moderate; no impact on Task, Change, or Problem tables.
8	**End-to-End Test**	Incidents assigned to imported users; manager details verified; Priority default confirmed.

Technology Stack

Component	Technology	Purpose
Data Source	Excel / CSV / XML	Structured employee data file for import
Import Engine	ServiceNow Import Sets	Native bulk data ingestion module
Staging Storage	Import Set Staging Table	Temporary data validation and holding area
Field Mapping	Transform Map	Maps source fields to sys_user table fields
Reference Resolution	Transform Map Script	Resolves Manager name to sys_user reference
User Repository	sys_user Table	ServiceNow's central user/employee record store
Form Enhancement	Form Designer / Dot Walking	Surfaces relational data on the Incident form
Config Override	Dictionary Override	Scoped Incident Priority default setting